

Day 10 - Integrating EOMs Numerically

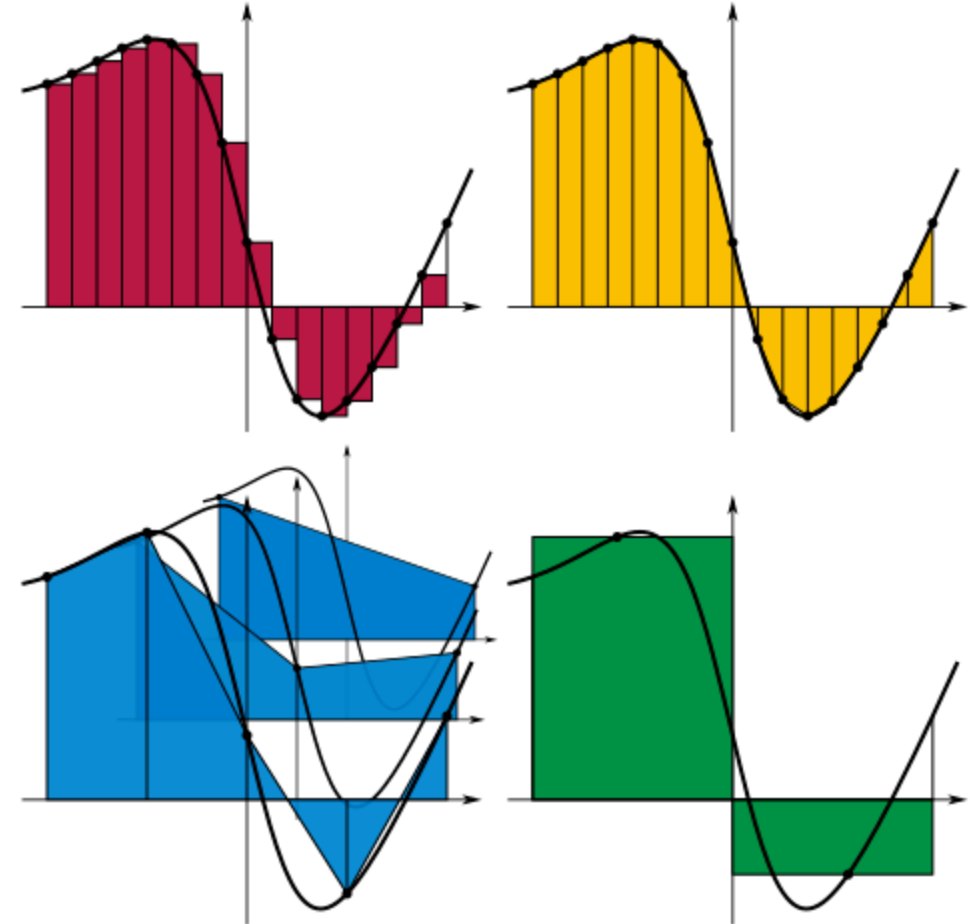
For Friday, please submit questions.



Scan to Join Session



Point your phone camera at this QR code to join the session



Announcements

- Homework 3 is posted (due Friday; late after Sunday)
- Homework 4 is posted (due next Friday; late after Sunday)
- **First midterm is coming up** (assigned 16 Feb)
 - One exercise will ask you to get started on your final project planning.
 - *Who are you gonna work with? What are you interested in studying?*
Start thinking about this!
- Limiting using [RaiseMyHand](#) to Workshop Days on (Fridays) only.
 - *Will start posting the link to collect your questions for those days.*

Weekly Seminars

WEDNESDAY, February 4, 2026

Astronomy Seminar, 1:30 p.m., BPS 1400 & Zoom

Speaker: Marvin Morgan, UCSB

Title: TBA

Zoom Link: <https://msu.zoom.us/j/93334479606?pwd=OtIXPWWhRPBfzYu53sl3trSJlaBYI7C.1>

Passcode: 825824

Weekly Seminars

THURSDAY, February 5, 2026

Colloquium, Seminar, 3:30 p.m., BPS 1415 & Zoom

Refreshments at 3:00 BPS in BPS 1400

Speaker: Glennys Farrar, New York University

Title: Origin of ultrahigh-energy cosmic rays in Binary Neutron Star Collisions
and the crucial roles of Nuclear Physics

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002

Weekly Seminars

FRIDAY, February 6, 2026

QuIC, Seminar, 12:40 p.m., BPS 1300 & Zoom

Speaker: Gregory Quiroz, APL

Title: Classical Non-Markovian Noise in Symmetry-Preserving Quantum Dynamics

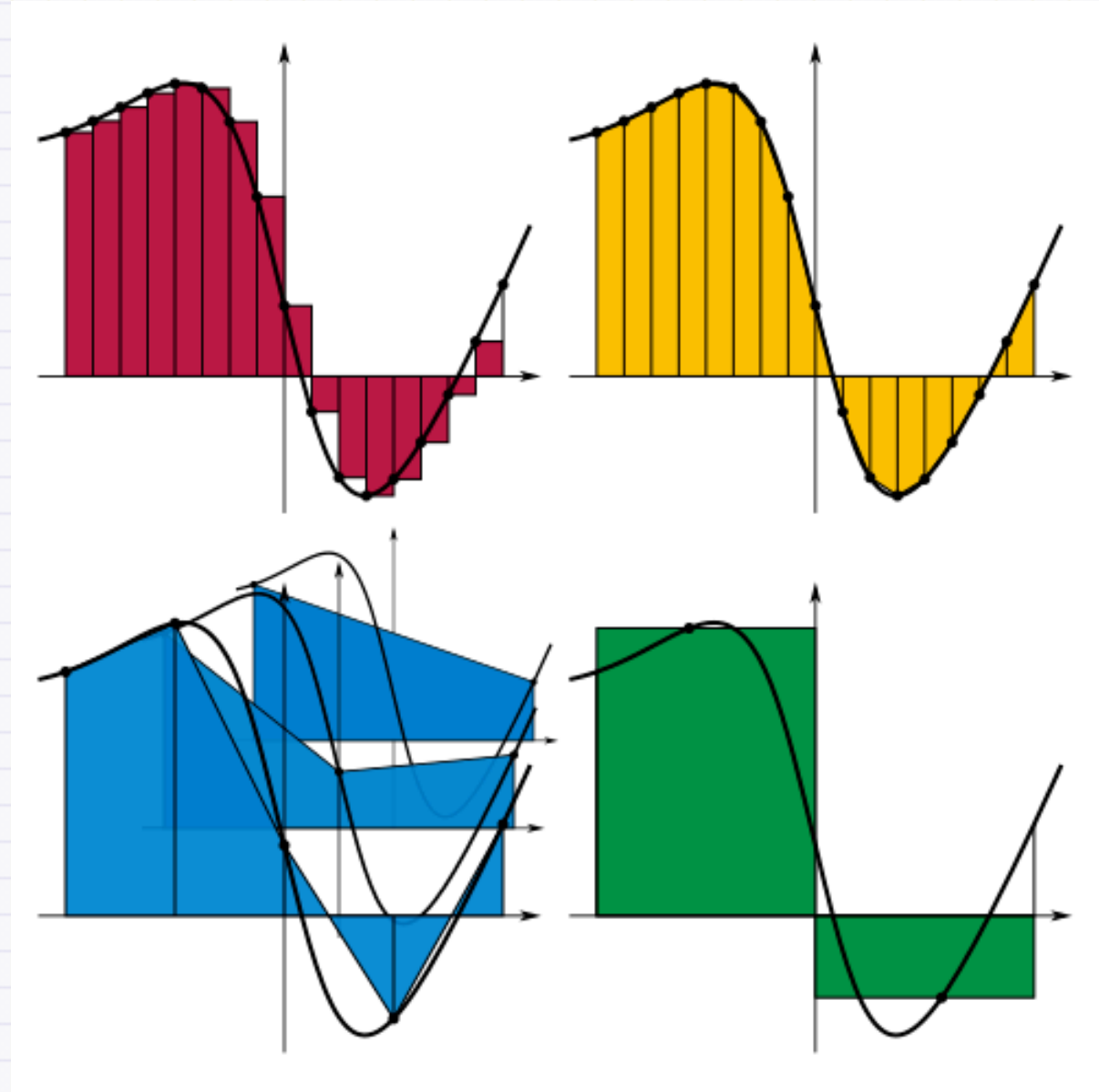
*For the full schedule, please see: <https://sites.google.com/msu.edu/quic-seminar/> or for more information, please reach out to Ryan LaRosa directly

Reminder:

Email Danny (caball14@msu.edu) your extra credit seminar write-ups

Goals for this week

- Establish a model for drag forces
- Develop an understanding of the process for modeling forces
- Produce equations of motion that can be investigated
- Start probing the behavior of these systems with math and computing



Model-to-EOM Pipeline for Classical Mechanics

1.  Develop conceptual description of the system; make justifiable assumptions
2.  Using a framework of physics (i.e., Newton's Laws, Lagrangian Dynamics), develop a mathematical model of the system
3.  Produce the equations of motion (EOM) by following the framework (i.e., ordinary differential equations)
4.  **Solve for trajectories of the system** (e.g., $x(t)$, $v(t)$, $v(x)$)

Most EOMs are nonlinear

We need approximate methods to produce trajectories.

Clicker Question 10-1

We found that the equation of motion for the spring-mass system was:

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

Your friends have proposed the following **particular solutions**:

1. $x(t) = A \cos(\omega t)$ 2. $x(t) = B \sin(\omega t)$ 3. $x(t) = A \cos(\omega t) + B \sin(\omega t)$
4. $x(t) = A \cos(\omega t + \phi)$ 5. $x(t) = B \sin(\omega t + \phi)$ 6. $x(t) = A \cos(\omega t + \phi) + B \sin(\omega t + \phi)$

How many of them are possible ok?

- (1) Only one (2) Two (3) Three
(4) Four (5) All of them

Should we start with general solutions or particular solutions?

Clicker Question 10-2

To convert the second order ODE for a spring mass system,

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

to two first order ODEs, we first introduce an equation for the velocity:

$$v = \dot{x}.$$

This is a first order ODE, and the first one we need. What is the other first order ODE?

Clicker Question 10-2

For spring-mass system ($\ddot{x} = -\omega^2 x$), and using the first order ODE that defines velocity $v = \dot{x}$, which other first order ODE can be used with the velocity equation to represent the spring mass system?

1. $\dot{x} = -\omega^2 x$
2. $\dot{v} = -\omega^2 x$
3. $\ddot{x} = a$
4. $\dot{v} = a$
5. Something else

Clicker Question 10-3

We derived the Euler step for constant acceleration:

$$v[i + 1] = v[i] + a * \Delta t$$
$$x[i + 1] = x[i] + v[i] * \Delta t$$

We can implement in Python for arrays `x` and `v` with:

```
v[i+1]=v[i]+a*deltat      # Constant Acceleration  
x[i+1]=x[i]+v[i]*deltat   # Euler Step
```

How can we implement a non-constant acceleration?

Clicker Question 10-3

How can we implement a non-constant acceleration?

```
v[i+1]=v[i]+a*deltat      # Constant Acceleration  
x[i+1]=x[i]+v[i]*deltat  # Euler Step
```

1. Add in `a[i]` to the first equation
2. Pre-compute the array `a` outside the loop
3. Compute `a[i]` in the loop
4. Change `v[i]` to `v[i+1]`
5. More than one of these

Numerical Integration Techniques for EOMs

Euler Method (Forward Euler)

- Simple, but not very accurate
- Does not conserve energy in many systems (especially oscillatory systems)

$$v[i + 1] = v[i] + a[i] * \Delta t$$

$$x[i + 1] = x[i] + v[i] * \Delta t$$

Notice that we use the acceleration at the beginning of the time step.

Numerical Integration Techniques for EOMs

Euler-Cromer Method (Semi-Implicit Euler)

- Simple, more accurate than Euler
- Conserves energy in many oscillatory systems

$$v[i + 1] = v[i] + a[i] * \Delta t$$
$$x[i + 1] = x[i] + v[i + 1] * \Delta t$$

Notice that we use the updated velocity at the end of the time step.

Numerical Integration Techniques for EOMs

Velocity Verlet Method

- More accurate than Euler and Euler-Cromer
- Conserves energy in many oscillatory systems

$$x[i + 1] = x[i] + v[i] * \Delta t + \frac{1}{2} a[i] * \Delta t^2$$
$$v[i + 1] = v[i] + \frac{a[i] + a[i + 1]}{2} * \Delta t$$

Notice that we use the average acceleration over the time step.

Be careful not to confuse this with your kinematic equations!

Other Common Numerical Integration Techniques for EOMs

- Runge-Kutta 2nd Order (RK2)
- Runge-Kutta 4th Order (RK4)
- Leapfrog Method
- Beeman's Algorithm
- ...

Many of these methods are more accurate, but also more complex to implement.

They are available in Python's scientific computing libraries (e.g., `scipy` .)